

Kai's Cookie Empire

Time limit per test: 2 seconds

Memory limit per test: 256 megabytes

Kai and Ethan are building a cookie empire.

Initially, you have g cookies. There are n types of workers you can hire. For each worker type i , hiring the worker costs a_i cookies, the worker produces b_i cookies per unit time after training, and the training process takes exactly t_i time units.

You can train at most one worker at a time. When you decide to hire a worker, you immediately pay a_i cookies, and the worker begins training. After exactly t_i time units, the worker finishes training and becomes active. Once active, the worker produces b_i cookies per unit time continuously until time T . Active workers do not occupy the training slot, and multiple active workers produce cookies simultaneously.

You may hire workers sequentially, including multiple workers of the same type. However, you must always have enough cookies to pay the hiring cost at the moment you start training a worker.

You are given a total time limit T . Workers only produce cookies after they finish training, and all production stops at time T .

Your goal is to maximize the total number of cookies you have at time T .

Input

The first line contains three integers n , g , and T ($1 \leq n \leq 2 \cdot 10^5$, $0 \leq g \leq 10^{18}$, $1 \leq T \leq 2 \cdot 10^5$). Each of the next n lines contains three integers a_i , b_i , and t_i ($1 \leq a_i, b_i \leq 10^{18}$, $1 \leq t_i \leq 10^9$).

Output

Print a single integer — the maximum number of cookies you can have at time T .

Examples

Input

```
3 10 10
5 3 2
8 5 3
4 2 1
```

Output

```
49
```

Input

```
2 0 5
10 100 3
5 1 1
```

Output

```
0
```

Note

In the first example, one optimal strategy is to first train the third worker, which costs 4 cookies and finishes at time 1, and then train the first worker, which costs 5 cookies and finishes at time 3. The workers then produce cookies for the remaining time, maximizing the final total.